

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0586

Conowingo 138 kV
February 03, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0586. The Interconnection Customer proposes a 79.3 MW solar photovoltaic facility interconnecting at the Conowingo 138 kV substation in the Maryland area. The request was received on February 27, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 312 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$100,146,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Springville - Alexander St 138 kV	Bixby Hill - Tombstone 500 kV	251.0	222	13.5
Springille - Conowingo Substation 138 kV	Tombstone - Bixby Hill Substation 500 kV	250.4	360	24.8
Glasgow - Dell Creek 138 kV	Bixby Hill - Tombstone 500 kV	234.7	180	5.4
Mill Cteek - Conoco S 138 kV	Alexander St - Springville Substation 138 kV	219.2	179	17.2
Mill Cteek Substation - Southpoint 138 kV line	Blooming Prairie - Mill Cteek 138 kV	219.1	172	5.3
Glasgow - BloomingPrairie 138 kV	Bixy Hill - Tombstone 500 kV	216.9	209	5.4
Mill Cteek - Blooming Prairie 138 kV	Tombstone - Bixby Hill 500 kV	216.0	214	5.4
Denhm Springs - Alexander St 138 kV	Bixby Hill - Tombstone 500 kV	205.2	190	10.4
Conoco South - Perryville 138 kV	Springville - Alexander St 138 kV	204.9	199	17.2
Springville - Waste Management Northern Oaks Lfgte 138 kV	Springville - Alexander St 138 kV	200.7	217	17.2
Perryville - Waste Management Northern Oaks Lfgte 138 kV	Alexander St - Springville 138 kV	200.4	212	17.2
Southpoint - Elizabeth Lake 138 kV Line	Blooming Prairie - Mill Cteek 138 kV	167.5	206	5.3
Star Peak - Elizabeth Lake 138 kV	Blooming Prairie - Mill Cteek 138 kV	139.9	224	5.3
Denham Springs - Otay Tap 138 kV Line	Star Peak 345 kV/138 kV TX	126.7	235	5.9
Shred - Otay Tap 138 kV Line	Star Peak 345 kV/138 kV Autotransformer	121.4	220	5.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Seminole - Shred 138 kV	Star Peak Substation 345/138 kV XFMR	118.9	200	5.9
Dell Creek - Keeney Substation 138 kV	Conowingo 345 kV/138 kV Transformer Bank	103.8	196	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Alexander Street - Springville 138 kV L	Rebuild line with higher-capacity conductor	\$5,184,000	100% (\$5,184,000)
Springille - Conowingo Substation 138 kV	Rebuild line with higher-capacity conductor	\$10,893,000	78% (\$8,456,000)
Dell Crk - Glasgow 138 kV	Rebuild line with higher-capacity conductor	\$1,822,000	100% (\$1,822,000)
Mill Cteek - Conoco S 138 kV	Rebuild line with higher-capacity conductor	\$16,789,000	100% (\$16,789,000)
Mill Cteek Substation - Southpoint 138 kV line	Rebuild line with higher-capacity conductor	\$4,333,000	100% (\$4,333,000)
Glasgow - BloomingPrairie 138 kV	Rebuild line with higher-capacity conductor	\$17,182,000	100% (\$17,182,000)
Mill Cteek - Blooming Prairie 138 kV	Rebuild line with higher-capacity conductor	\$5,559,000	100% (\$5,559,000)
Denhm Springs - Alexander St 138 kV	Rebuild line with higher-capacity conductor	\$14,562,000	100% (\$14,562,000)
Perryville - Conoco South 138 kV L	Rebuild line with higher-capacity conductor	\$4,138,000	100% (\$4,138,000)
Springville - Waste Management Northern Oaks Lfgte 138 kV Line	Rebuild line with higher-capacity conductor	\$2,725,000	100% (\$2,725,000)
Perryville - Waste Management Northern Oaks Lfgte 138 kV	Rebuild line with higher-capacity conductor	\$5,266,000	100% (\$5,266,000)
Southpoint - Elizabeth Lake 138 kV Line	Rebuild line with higher-capacity conductor	\$1,539,000	100% (\$1,539,000)
ElizabethLake - Star Peak 138 kV	Rebuild line with higher-capacity conductor	\$2,293,000	100% (\$2,293,000)
Denhm Springs - Otay Tap 138 kV line	Rebuild line with higher-capacity conductor	\$5,796,000	100% (\$5,796,000)
Otay Tap - Shred 138 kV	Rebuild line with higher-capacity conductor	\$2,068,000	100% (\$2,068,000)
Shred - Seminole 138 kV	Rebuild line with higher-capacity conductor	\$1,677,000	100% (\$1,677,000)
Dell Creek - Keeney Substation 138 kV	Reconductor line	\$757,000	100% (\$757,000)
	Total allocated to project		\$100,146,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.